



COEP Technological University (COEP Tech) Pune
A Unitary Public University of Govt of Maharashtra
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(Formerly College of Engineering, Pune)
Shivajinagar, Pune – 411005. INDIA.

Department of Civil Engineering

Tel: +91-20-25507200, Fax: +91-20-25507299

INTERNATIONAL STANDARDS – UNITWISE ASSIGNMENTS

Name of Student:

MIS No:

Branch:

UNIT -1

INTRODUCTION TO STANDARDIZATION & STANDARDIZATION BODIES WITH THEIR FRAMEWORKS

1. Discuss the objectives and benefits of standardization in global trade and technology.
2. Explain the structure and function of ISO (International Organization for Standardization).
3. Describe the lifecycle of a standard from development to implementation.
4. How do standardization bodies coordinate to prevent conflicting standards?
5. How do national standardization bodies (like BIS in India) align with international standards?

UNIT II

ISO STANDARDS OVERVIEW & CORE INTERNATIONAL STANDARDS

1. Explain the structure of an ISO standard document.
2. Compare ISO 9001 and ISO 27001 in terms of objectives and application.
3. What are the steps involved in getting ISO certified?
4. Describe the importance of ISO 27002 and how it complements ISO 27001.
5. Discuss the concept of continual improvement in ISO standards.

UNIT III

**STANDARDS DEVELOPMENT PROCESS OF
STANDARDIZATION**

Each Question carry 5 Marks

1. Stages of Standard Development

1. What are the main stages involved in the development of a standard?
2. Describe the activities involved in the proposal stage of standard development.
3. How is the drafting stage conducted in the development of standards?

2. Stakeholders in the Standardization Process

4. What role do government agencies play in standardization?
5. Why is the involvement of consumers important in the standards development process?
6. Explain the influence of industry associations in standardization.

3. National Participation and Harmonization with Global Standards

7. What is national participation in standardization, and why is it important?
8. What are the benefits of harmonizing national standards with international standards?
9. Describe the challenges faced during harmonization of standards across countries.

4. Technical Committees and Working Groups

10. How are Technical Committees organized within organizations such as ISO or IEC?
11. What are the typical deliverables produced by a Technical Committee?

5. Mapping Standards in Specific Subjects

12. What is ISO 128, and what area does it cover in standardization?
13. How do standard codes impact safety and quality in civil engineering projects?

6. From Proposal to Publication: Committee Roles, Drafts, Ballots

14. What roles do committee members play from the proposal stage to the publication of a standard?

15. What is a ballot in the context of standards development?

16. How does the ballot process ensure consensus among stakeholders?

7. Revision Cycles and Case Studies (e.g., ISO 128)

17. Why is it necessary to revise standards periodically?

18. Describe the revision cycle process for international standards.

19. How are stakeholders informed and involved during a standard's revision?

UNIT IV

APPLICATION IN INDUSTRY & COMPLIANCE

1. Implementing Standards in Product Design, Manufacturing, and Documentation

1. What are the key benefits of implementing standards in product design?
2. How do standards influence manufacturing processes?

2. Compliance Tools: Certificates, Audits, Conformity Assessment

6. What are the typical steps involved in obtaining product certification?
7. How do third-party conformity assessments enhance trust in products?

3. Use of Standards in Various Sectors: Manufacturing, IT, Health, Energy

8. How are standards applied in the manufacturing industry to ensure product quality?
9. Describe how energy sector standards contribute to sustainability and efficiency.
10. Why is sector-specific standardization important for safety and performance?

4. Quality Assurance and Standard Compliance

11. How do standards support quality assurance in organizations?
12. What is the relationship between quality management systems (like ISO 9001) and standard compliance?

5. Risk Management and Legal Implications

13. How do standards help in managing risks associated with products and processes?
14. Discuss the legal implications of failing to comply with industry standards.

6. Emerging Trends in Standardization: Digital Standards, Sustainability, etc.

15. What are digital standards, and why are they becoming important?
16. How is standardization contributing to sustainability goals in industries?
17. Describe the role of standards in the development and adoption of IoT (Internet of Things).